Agile Development Process and Devops (ISWE406L) -LAB FAT

Slot: C2+TC2

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Regno:23MIS0518

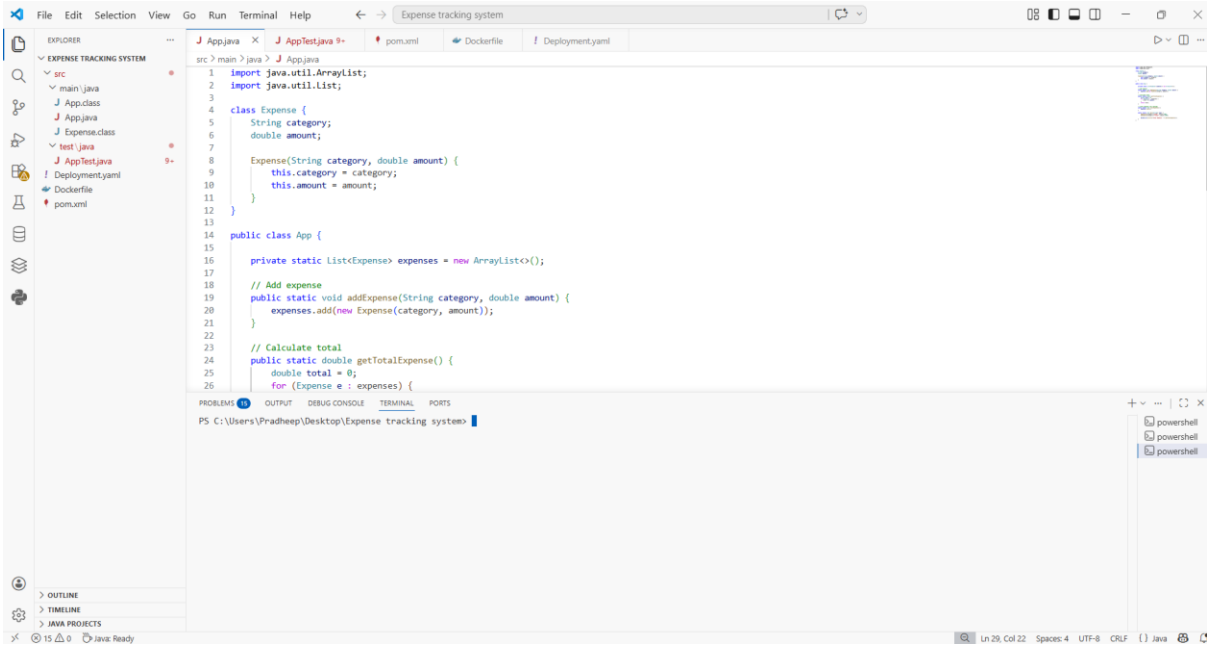
Set:6

Tasks:

1.Develop expense tracking logic

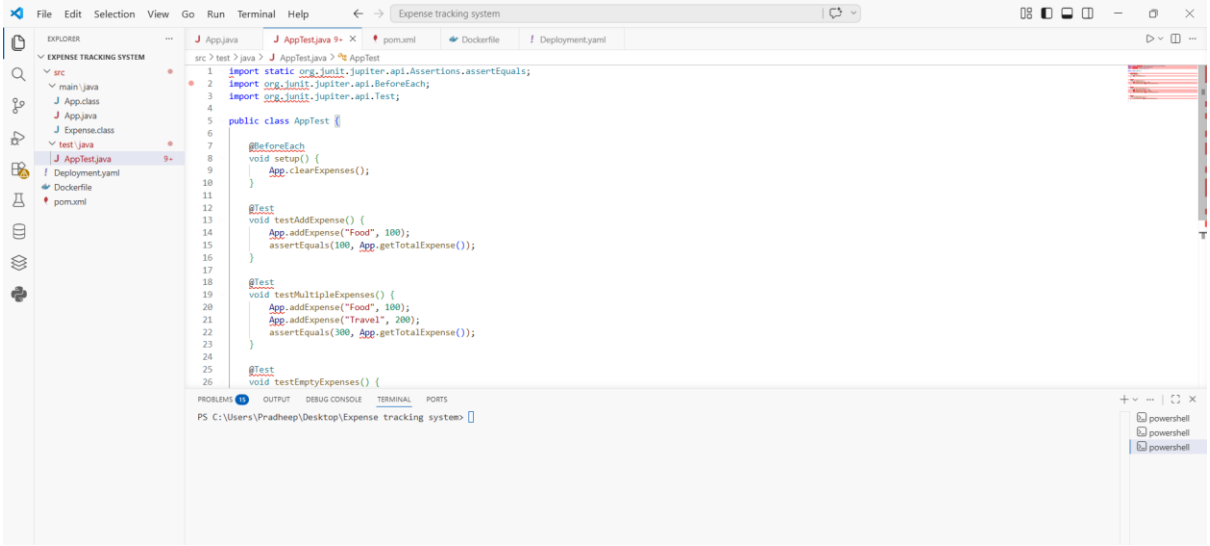
Creating files:

App.java

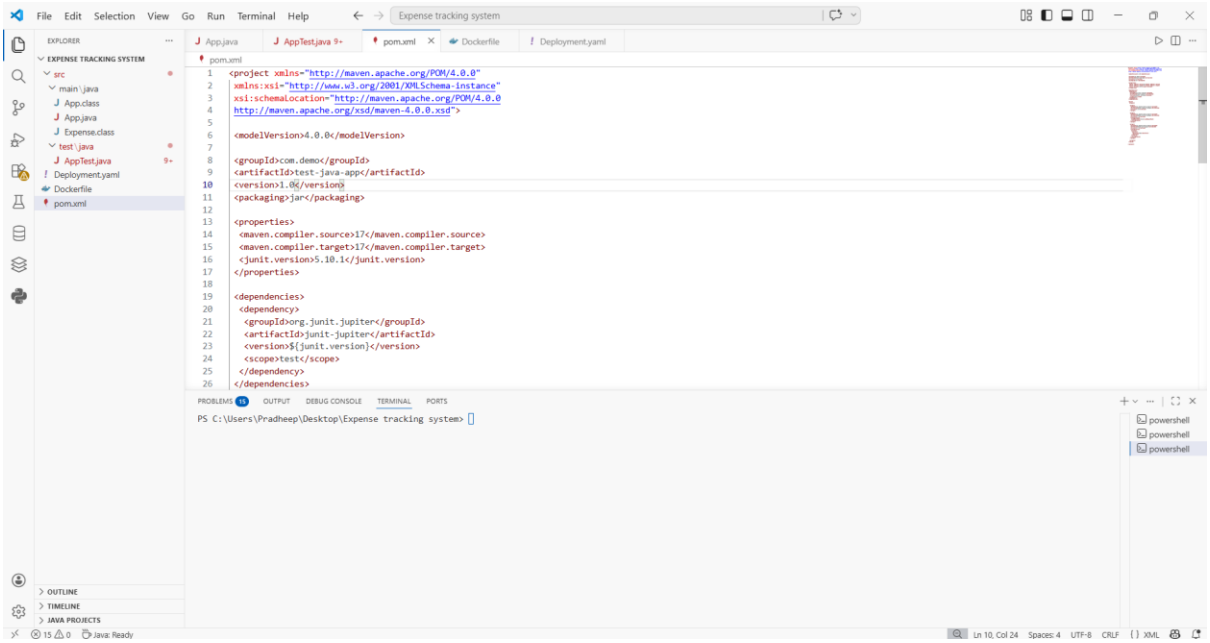


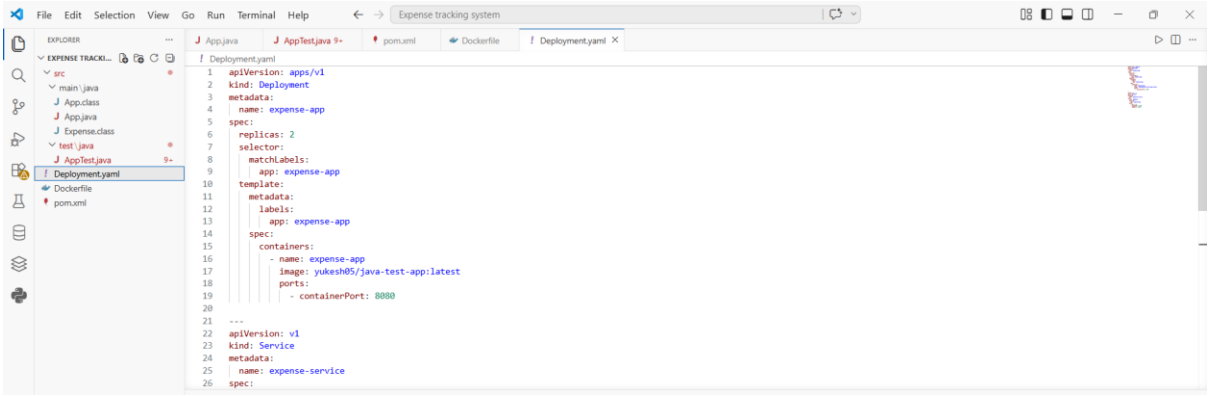
2.Write junit tests for calculations

Apptest.java



Pom.xml





Mvn clean test

```
PROBLEMS 15   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\Pradheep\Desktop\Expense tracking system> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\Expense tracking system\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\Expense tracking system\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.037 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.062 s
[INFO] Finished at: 2026-04-13T10:20:45+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Pradheep\Desktop\Expense tracking system>
```

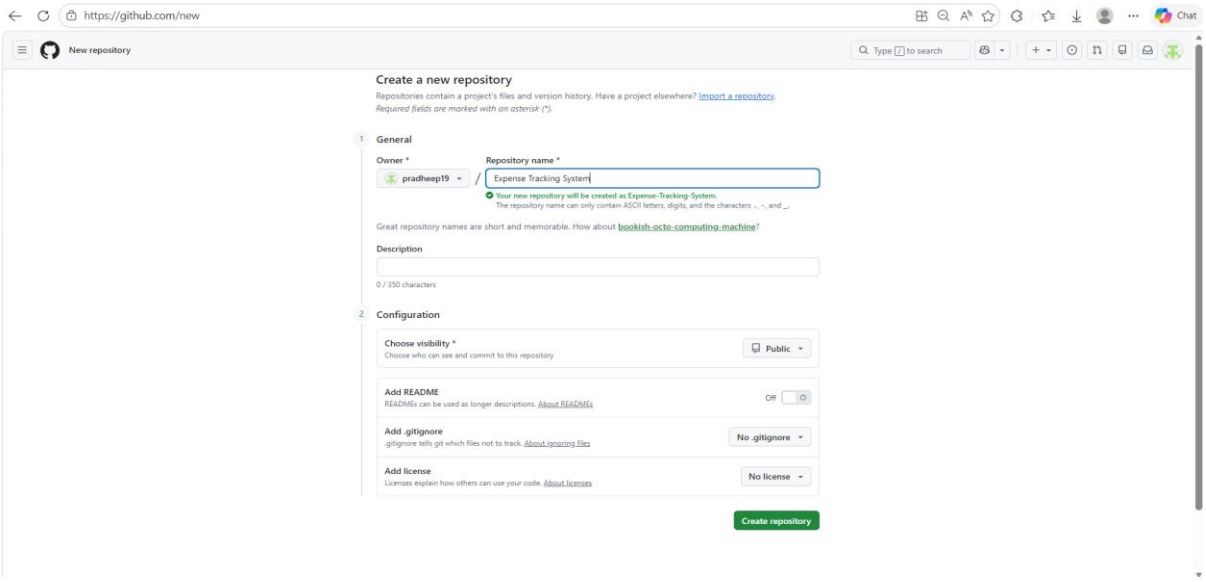
Mvn clean package:

```
PS C:\Users\Pradheep\Desktop\Expense tracking system> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\Pradheep\Desktop\Expense tracking system\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\Expense tracking system\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\Expense tracking system\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.042 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\Pradheep\Desktop\Expense tracking system\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.277 s
[INFO] Finished at: 2026-04-13T10:22:11+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Pradheep\Desktop\Expense tracking system>
```

Pusht code to Git:

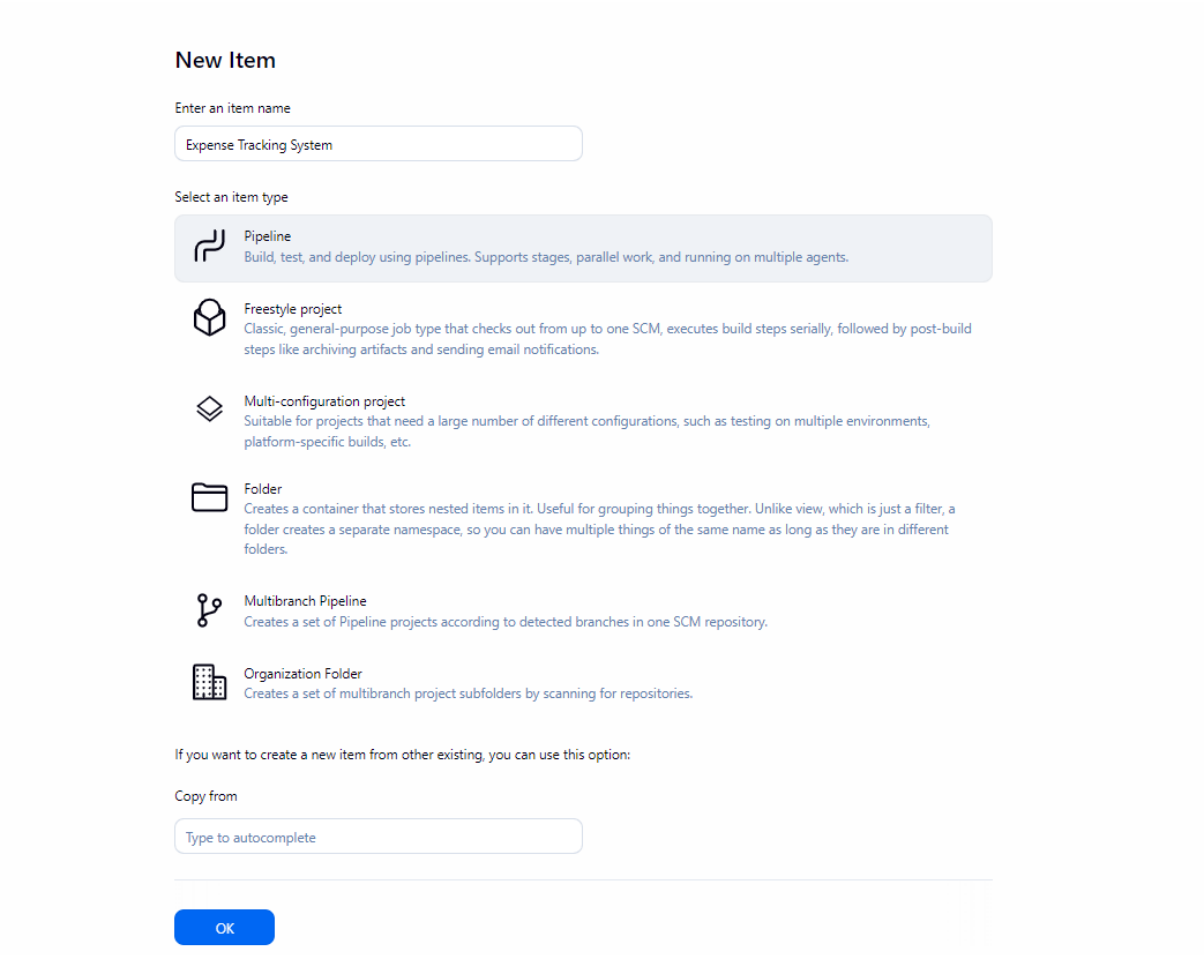
```
PS C:\Users\Pradheep\Desktop\Expense tracking system> git init
Initialized empty Git repository in C:/Users/Pradheep/Desktop/Expense tracking system/.git/
PS C:\Users\Pradheep\Desktop\Expense tracking system> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/surefire-reports/TEST-AppTest.xml', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\Pradheep\Desktop\Expense tracking system> git commit -m "Initial commit - Expense Tracking System with CI/CD"
[master (root-commit) de0dee2] Initial commit - Expense Tracking System with CI/CD
 18 files changed, 248 insertions(+)
 create mode 100644 Deployment.yaml
 create mode 100644 Dockerfile
 create mode 100644 pom.xml
 create mode 100644 src/main/java/App.class
 create mode 100644 src/main/java/App.java
 create mode 100644 src/main/java/Expense.class
 create mode 100644 src/test/java/AppTest.java
 create mode 100644 target/classes/App.class
 create mode 100644 target/classes/Expense.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/AppTest.txt
 create mode 100644 target/surefire-reports/TEST-AppTest.xml
 create mode 100644 target/test-classes/AppTest.class
 create mode 100644 target/test-java-app-1.0.jar
PS C:\Users\Pradheep\Desktop\Expense tracking system> git branch -M main
PS C:\Users\Pradheep\Desktop\Expense tracking system> git remote add origin https://github.com/pradheep19/Expense-Tracking-System
PS C:\Users\Pradheep\Desktop\Expense tracking system> git push -u origin main
Enumerating objects: 36, done.
Counting objects: 100% (36/36), done.
Delta compression using up to 16 threads
Compressing objects: 100% (26/26), done.
Writing objects: 100% (36/36), 9.62 KiB | 2.40 MiB/s, done.
Total 36 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/pradheep19/Expense-Tracking-System
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Pradheep\Desktop\Expense tracking system>
```

Create Repository:

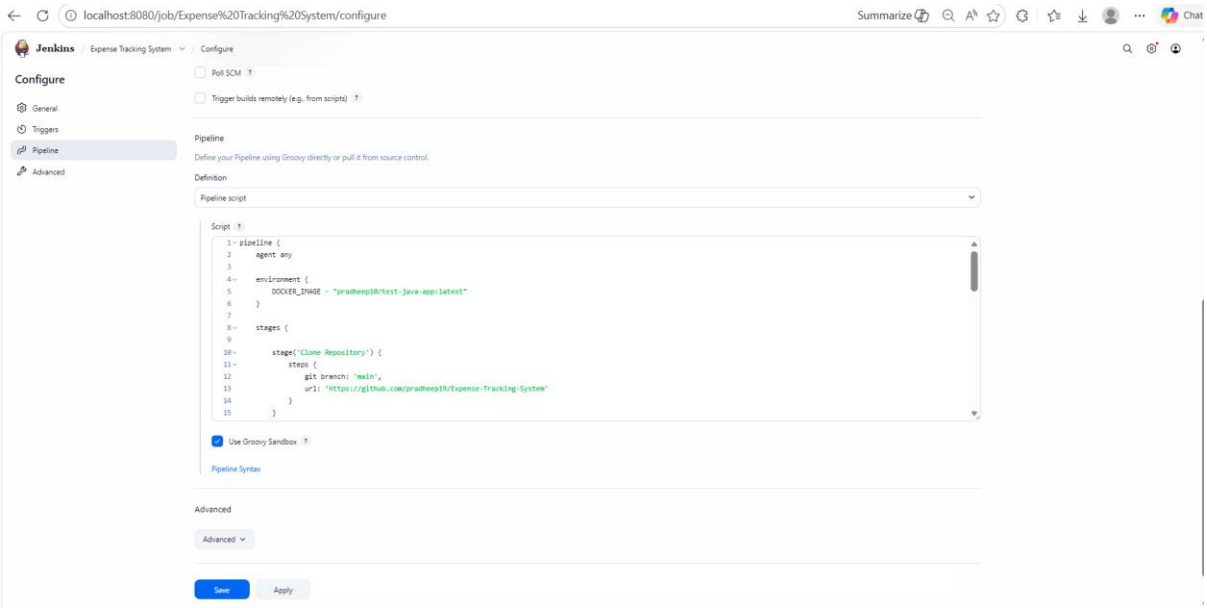


3.ImplementCI/CD pipeline

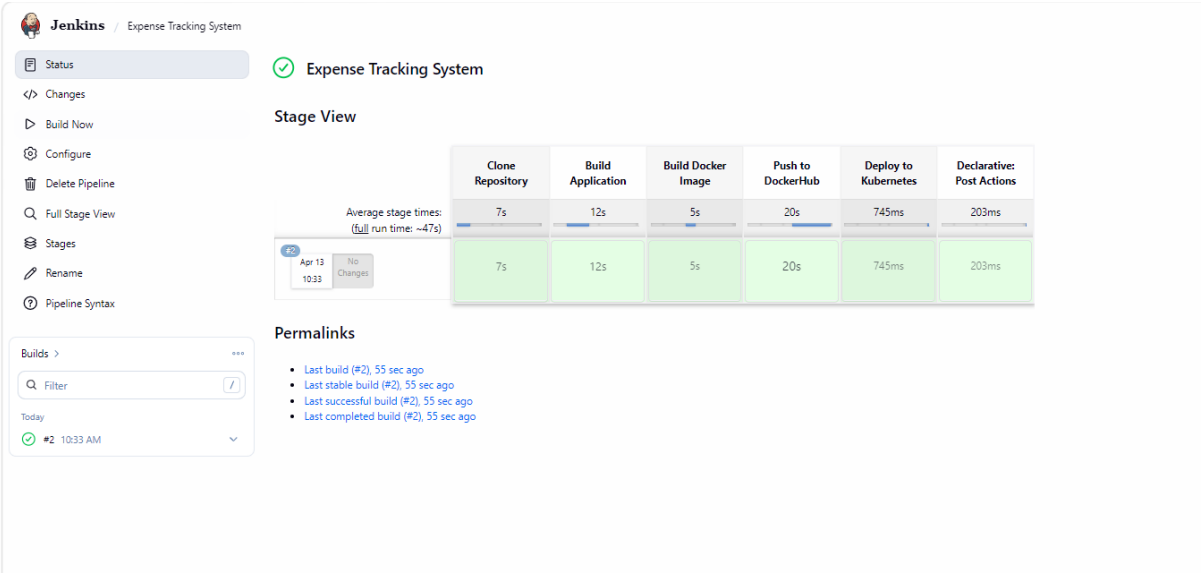
Create pipeline in Jenkin:



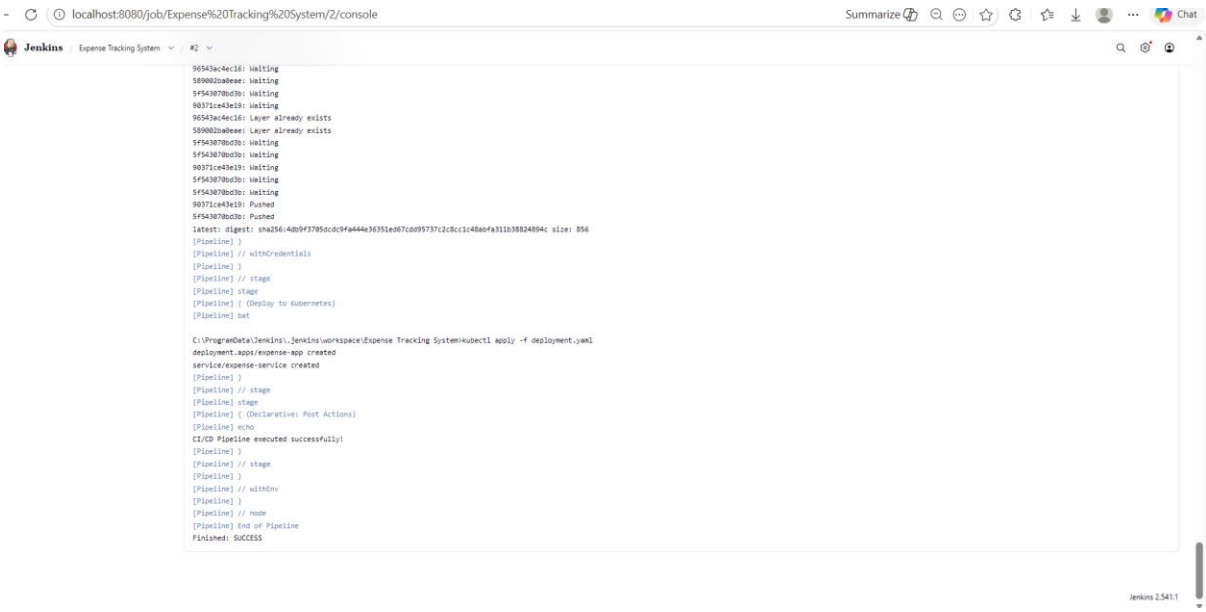
Jenkin configuration:



Automated build process:

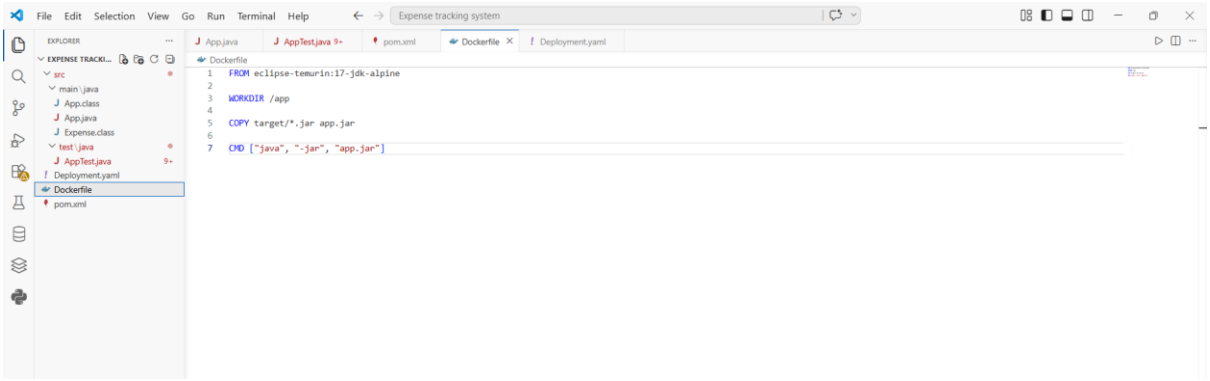


Console output

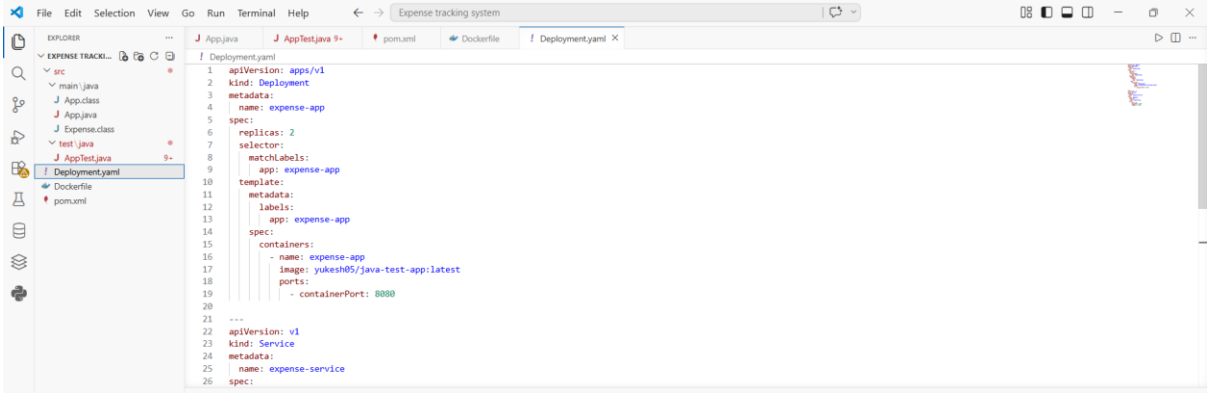


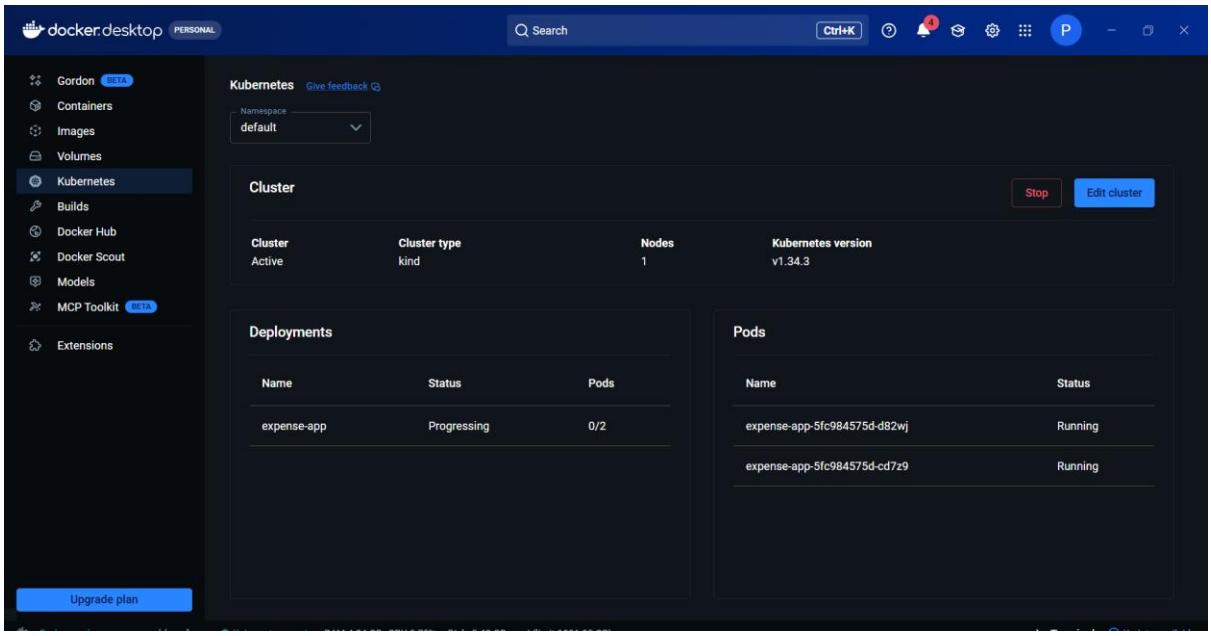
4.deploy services on Docker and Kubernetes:

Docker file:

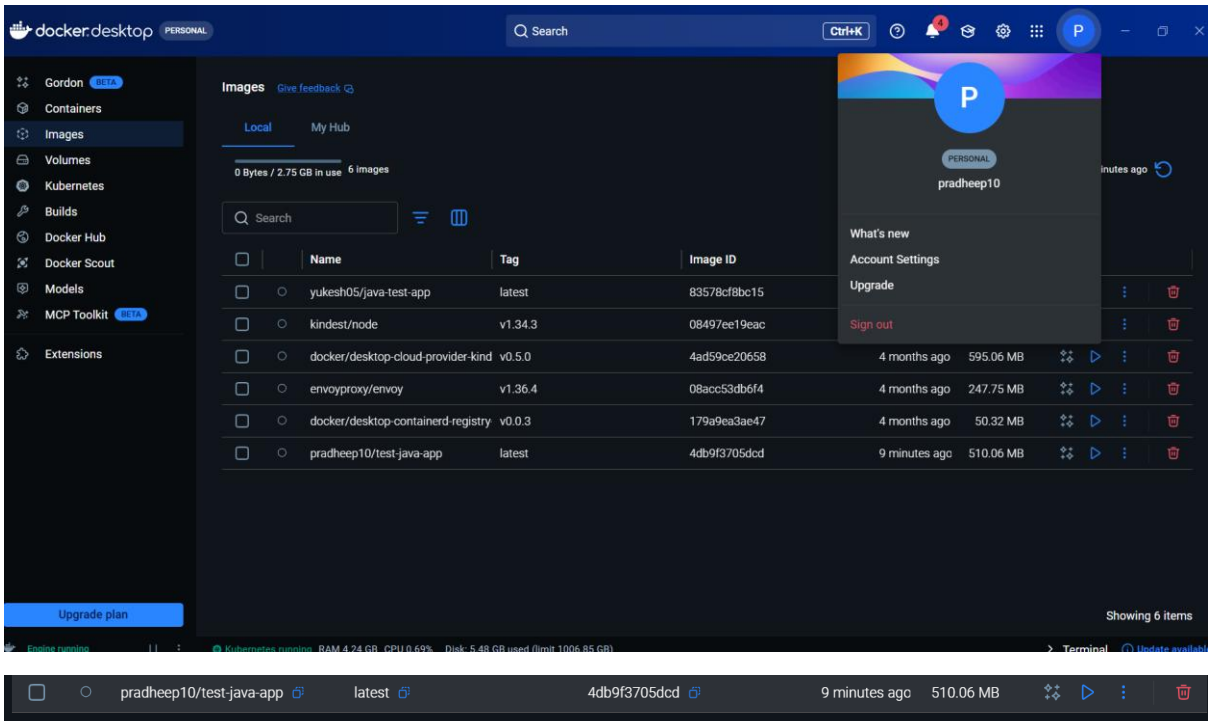


Deployment.yaml:





Docker Image:



```
Command Prompt

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Pradheep>kubectl get svc
NAME              TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
expense-service   NodePort    10.96.62.69   <none>        80:30007/TCP   10m
kubernetes        ClusterIP   10.96.0.1     <none>        443/TCP        14m

C:\Users\Pradheep>
```